SCOREBOARD – Hadamard 14408 Paley-I (2026-10-09)

Cutoff: 2026-10-09T19:27:17Z (~3:27PM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14408 Paley-I (q=14407)	PASS	0	3092.1554	**IN SPK** (Paley-I)
H_14408 Paley-II (q=7203)	FAIL/SKIP	–	–	q=7203=3*7? NOT prime power; ===3 mod 4 needs ===1
H_14408 Kronecker (4*3602)	FAIL/SKIP	–	–	3602===2 mod 4 NOT Hadamard-admissible; factors 2*1801

Units: 715/715 * row blocks 113/113 * overall_pass=True * wall_total=3131.4621s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*31163 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3^2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2^3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*37167 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3ˆ2*1571 NOT prime -> FAIL;**

Kronecker 14140=4*3535; 3535=57101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^257101 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=43555; 3555=3^2579 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3^2579 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7129->n=14260 Pal
NOT prime -> FAIL; Kronecker 14260=43565; 3565=5*2331 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^252331 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | next Paley-I DIRECT after 14408 (and FIRST any direct route after 14408): q=14411->n=14412 (prime ===3 mod 4; no other prime in 14408..14410; Paley-II q2=7205=5*11*131 NOT prime power (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14412=43603; 3603=3*1201 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3603); 14412=2^2*3*1201 ===12 mod 16) | further Paley-I DIRECT after 14412: 14420 (q=14419), 14424 (q=14423) | FIRST Paley-II DIRECT after 14408: q2=7213->n=14428 (q2=7213 prime ===1 mod 4; Paley-I q=14427=3^27229 NOT prime -> FAIL; Kronecker 14428=43607; 3607 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3607); 14428=2^23607 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII); comes after Paley-I 14412/14420/14424) | Kronecker-only (note-only, NOT on box): 14416=43604 (H_3604 via Paley-II q2=1801 prime ===1 mod 4; 14416 itself has no direct Paley route: 14415=3*5*31^2 composite, 7207 prime ===3 mod 4) (or gap 2052 / Track B)